IMPLEMENTATION AND EXPERIMENT REPORT

Particle filter localization and EKF SLAM

This report covers the coding components of **Questions 2 and 3(b)**, including the required experiment figures and discussions. The separate written-solutions document covers Questions 1 and 3(a). Results here were obtained by running the supplied datasets; no synthetic results are presented as homework experiments.

Question 2(a): prediction. Each particle samples the three independent Gaussian noise terms in the velocity model. The six supplied alpha coefficients define variances, so normal sampling uses their square roots. A sinc-based expression handles exactly zero angular velocity without division by zero. Headings wrap to [-pi, pi). Occupied endpoints are rejected; after ten failed proposals the particle retains its previous pose, following the starter's suggestion.

Question 2(b): observation model. Exact grid traversal predicts beam ranges. The scan likelihood is the independent-beam product of the hit, short, maximum-range and random components. The hit Gaussian and short exponential are normalized on their permitted intervals. The maximum-range term is the textbook point mass with a numerical saturation tolerance. All likelihood arithmetic used by the filter is in log space.

Question 2(c): recursion. Bootstrap importance weights multiply the previous weights by the new scan likelihood. This recurrence is retained when resampling is skipped. Systematic resampling draws N particles with replacement and resets weights to 1/N. Position is a weighted mean; orientation is a circular mean. ESS = 1 / sum(normalized_weight squared).

Setting	Value in reported PF runs
Random seed / beam sparsity	7 / every fifth beam (9 of 41 beams)
Known-start initialization	Gaussian xy, sigma = 0.5 m; uniform heading, as in starter
Unknown-start initialization	Uniform over free map area and [-pi, pi) heading
Hit / short / max / random	0.97 / 0.01 / 0.01 / 0.01
Hit sigma / short rate / max range	0.5 m / 1 per m / 20 m
Adaptive resampling	Resample when ESS < 2N/3

Evaluation. PF input column k produces estimate k+1, matched to XGT column k+1. Ground truth is used only for evaluation and figures. RMSE is the square root of the mean squared Euclidean position error over the trajectory. Final cloud spread is the square root of the weighted xy covariance trace. Path2 files have no ground truth, so scan consistency and concentration cannot establish absolute localization accuracy.

QUESTIONS 2(D) AND 2(E)

Particle-count sweep

All entries use always-resampling and seed 7. Asterisks mark the selected baseline counts. These are experimentally chosen counts, not universal minimum requirements.

Known start	N	RMSE (m)	Final error (m)	Spread (m)
simple path1	10	4.028	6.334	0.000
simple path1	100	0.152	0.054	0.096
simple path1	1000 *	0.142	0.057	0.078
simple path3	10	4.899	0.791	0.004
simple path3	100	0.150	0.202	0.093
simple path3	1000 *	0.099	0.121	0.098
loops path0	10	10.584	11.867	0.002
loops path0	100	0.091	0.057	0.080
loops path0	1000 *	0.057	0.085	0.085
loops path1	1	7.514	2.087	0.000
loops path1	10	0.180	0.154	0.024
loops path1	100	0.061	0.052	0.071
loops path1	1000 *	0.045	0.023	0.083
rooms path1	10	5.395	7.705	0.007
rooms path1	100	0.091	0.068	0.028
rooms path1	1000 *	0.062	0.061	0.047

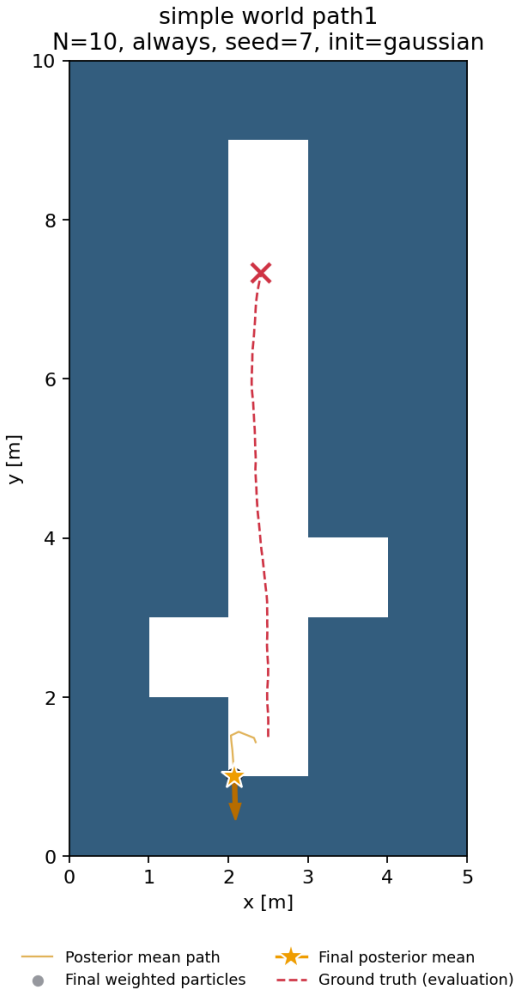
Unknown start	N	Final scan log L	Spread (m)	ESS / N
simple path2	250	-7.811	0.028	0.765
simple path2	2000	-7.985	0.018	0.763
simple path2	5000 *	-1.672	0.101	0.827
loops path2	250	-2.383	0.067	0.915
loops path2	2000 *	-2.397	0.073	0.919
loops path2	5000	-2.388	0.073	0.915
rooms path2	250	-36.740	0.013	0.975
rooms path2	2000 *	-2.151	0.048	0.956
rooms path2	5000	-7.095	0.036	0.208

Final scan log L is evaluated at the posterior mean with the same beam model; higher is better for that same scan. It is a diagnostic, not a ground-truth error. Large ESS and a narrow cloud can also occur after the filter has retained the wrong hypothesis.

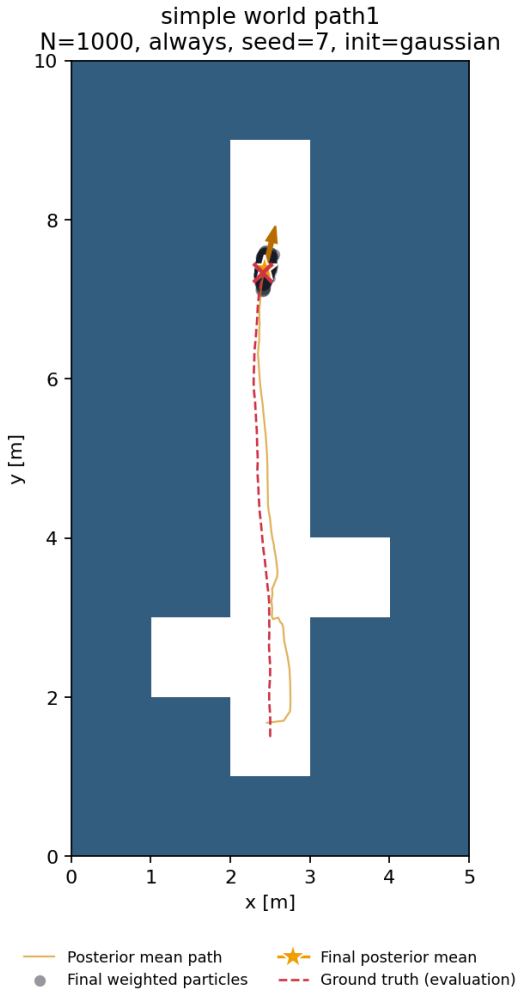
QUESTION 2(D): KNOWN INITIAL POSITION

simple world path1

Run	Position RMSE	Final position error	Heading RMSE
Failure example: N=10	4.028 m	6.334 m	175.11 deg
Selected: N=1000	0.142 m	0.057 m	2.56 deg



Failure example, N=10, seed 7.



Selected result, N=1000, seed 7.

The ten-particle run retains an incorrect heading hypothesis and stalls near the lower obstacle boundary. Its tiny final cloud is overconfident: the pose is far from ground truth. The 1000-particle run follows the observed path closely.

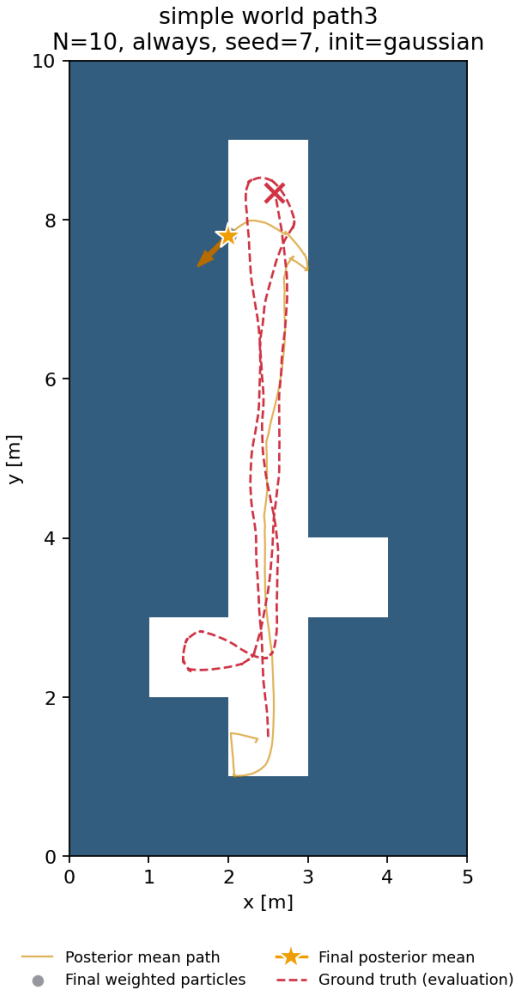
Black points show the final weighted particle population; the orange star is its weighted mean. The dashed red path and red endpoint are ground truth used only for evaluation.

Source: results/pf/simple_world_path1_N*_always_s7/ (JSON metrics, PNG figures and NPZ histories).

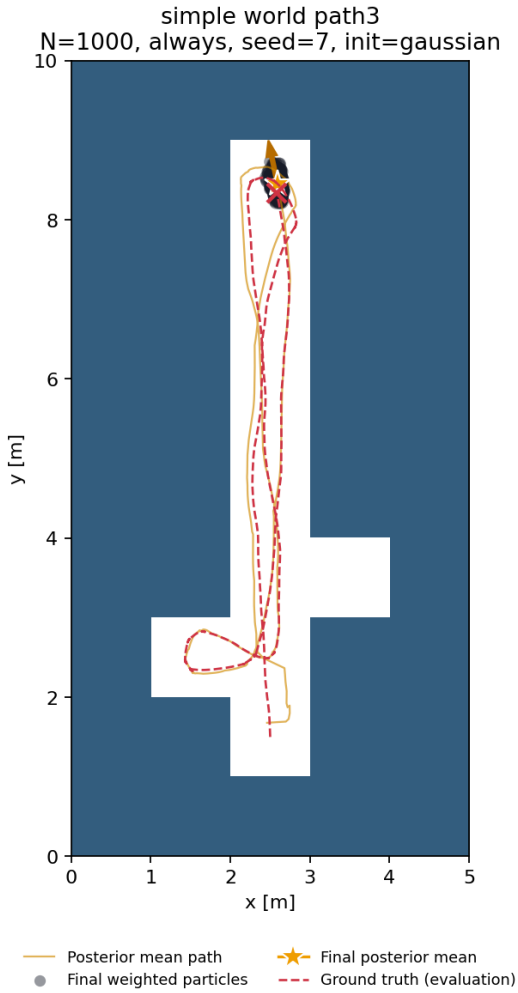
QUESTION 2(D): KNOWN INITIAL POSITION

simple world path3

Run	Position RMSE	Final position error	Heading RMSE
Failure example: N=10	4.899 m	0.791 m	161.20 deg
Selected: N=1000	0.099 m	0.121 m	2.27 deg



Failure example, N=10, seed 7.



Selected result, N=1000, seed 7.

Ten particles produce a poor trajectory and heading estimate. The final position happens to be relatively near the endpoint, but its smaller final error does not erase the large trajectory RMSE. A sufficient cloud is needed throughout the trajectory.

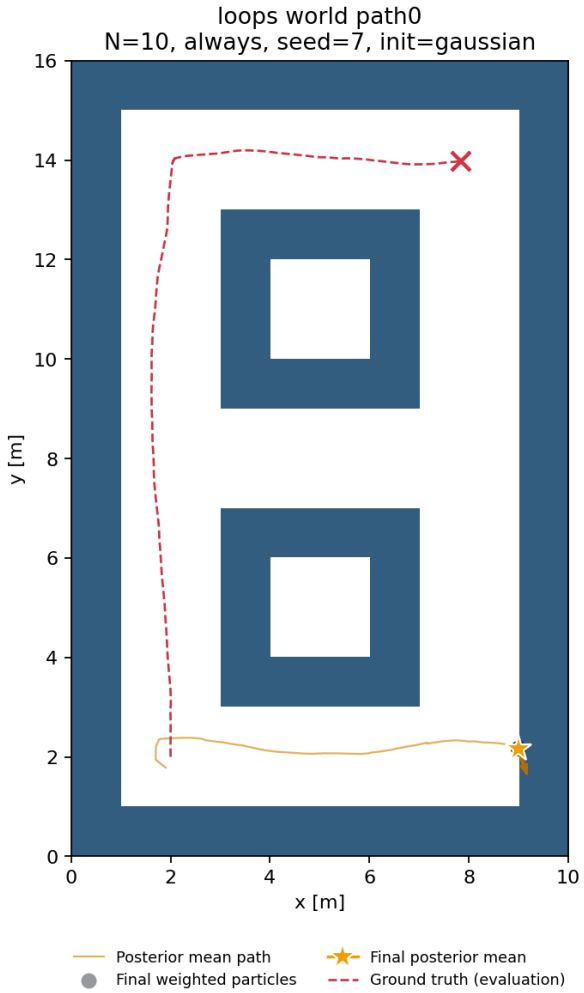
Black points show the final weighted particle population; the orange star is its weighted mean. The dashed red path and red endpoint are ground truth used only for evaluation.

Source: results/pf/simple_world_path3_N*_always_s7/ (JSON metrics, PNG figures and NPZ histories).

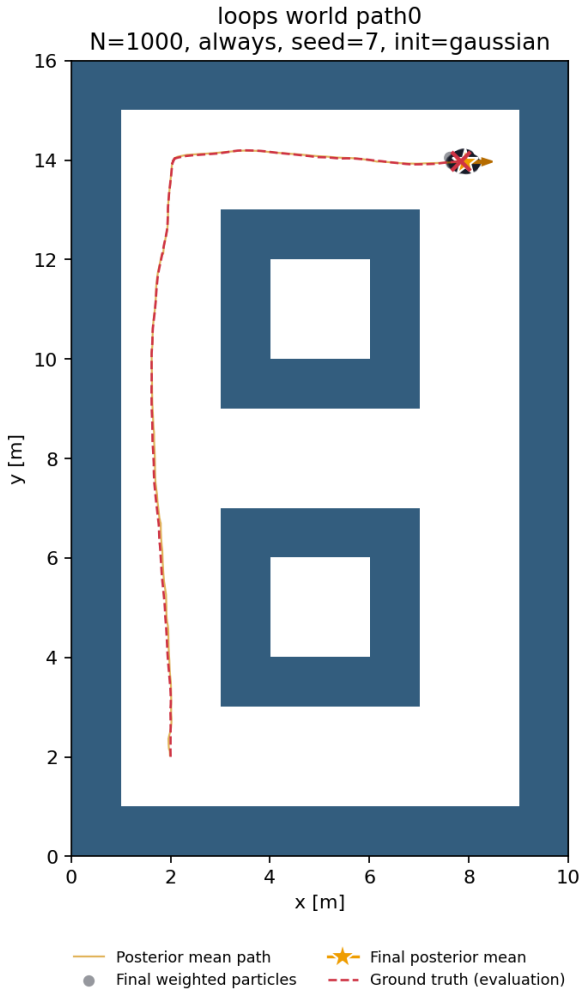
QUESTION 2(D): KNOWN INITIAL POSITION

loops world path0

Run	Position RMSE	Final position error	Heading RMSE
Failure example: N=10	10.584 m	11.867 m	87.65 deg
Selected: N=1000	0.057 m	0.085 m	1.80 deg



Failure example, N=10, seed 7.



Selected result, N=1000, seed 7.

The ten-particle run finishes in the wrong corridor. Its final effective sample size is almost ten even though its position error is large. ESS measures weight concentration among surviving particles, not correctness of their locations.

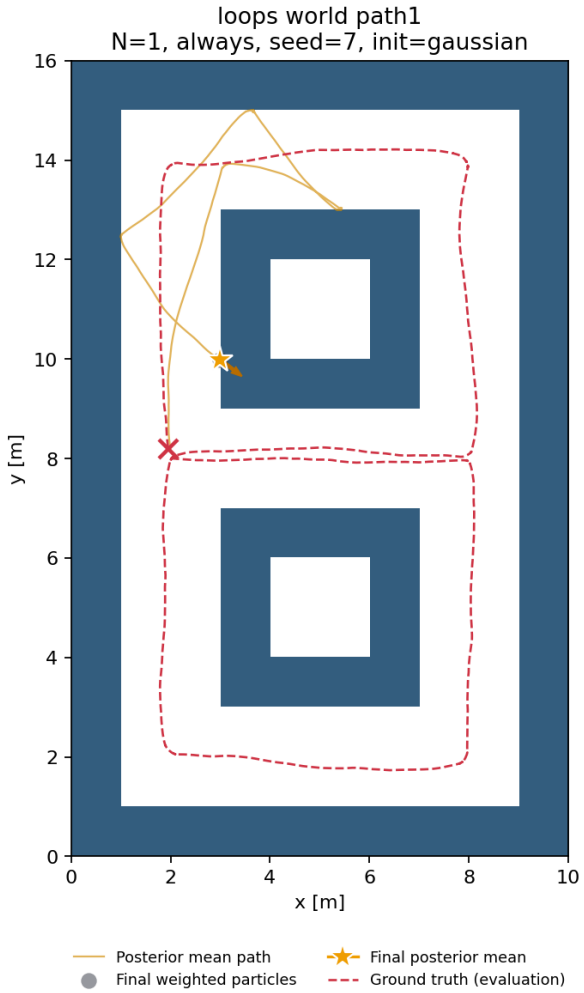
Black points show the final weighted particle population; the orange star is its weighted mean. The dashed red path and red endpoint are ground truth used only for evaluation.

Source: results/pf/loops_world_path0_N*_always_s7/ (JSON metrics, PNG figures and NPZ histories).

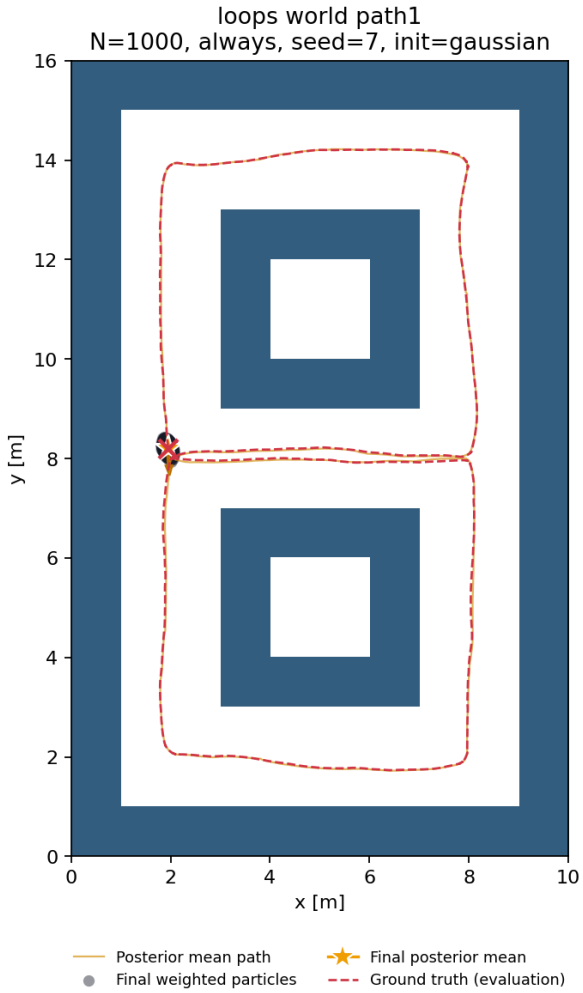
QUESTION 2(D): KNOWN INITIAL POSITION

loops world path1

Run	Position RMSE	Final position error	Heading RMSE
Failure example: N=1	7.514 m	2.087 m	52.64 deg
Selected: N=1000	0.045 m	0.023 m	1.95 deg



Failure example, N=1, seed 7.



Selected result, N=1000, seed 7.

Ten particles actually work reasonably well in this particular run; they are not labeled a failure. The added one-particle experiment fails. With one particle, normalized weighting is always one and resampling cannot create a competing pose hypothesis.

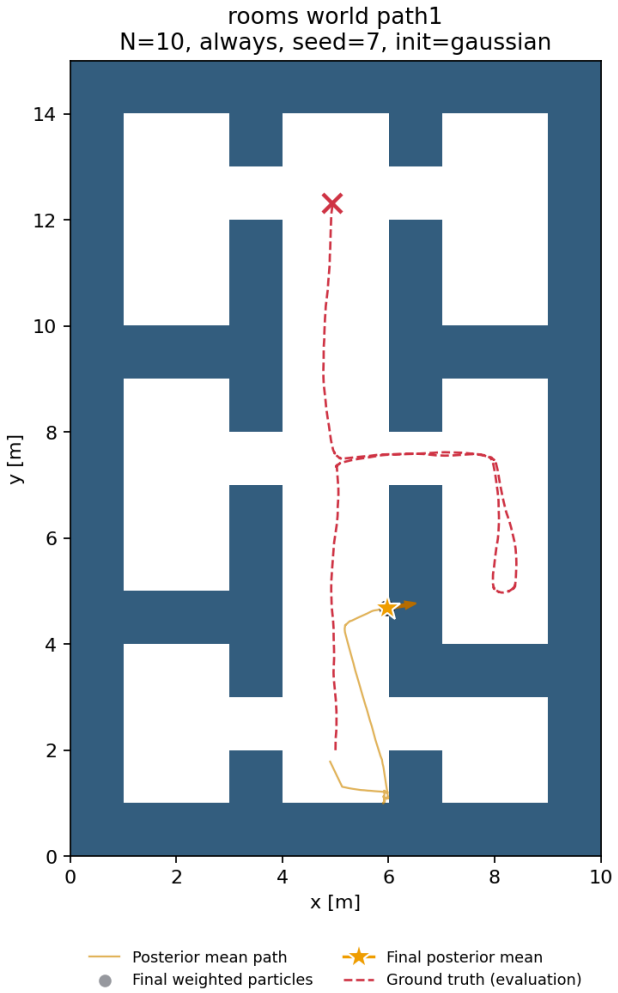
Black points show the final weighted particle population; the orange star is its weighted mean. The dashed red path and red endpoint are ground truth used only for evaluation.

Source: results/pf/loops_world_path1_N*_always_s7/ (JSON metrics, PNG figures and NPZ histories).

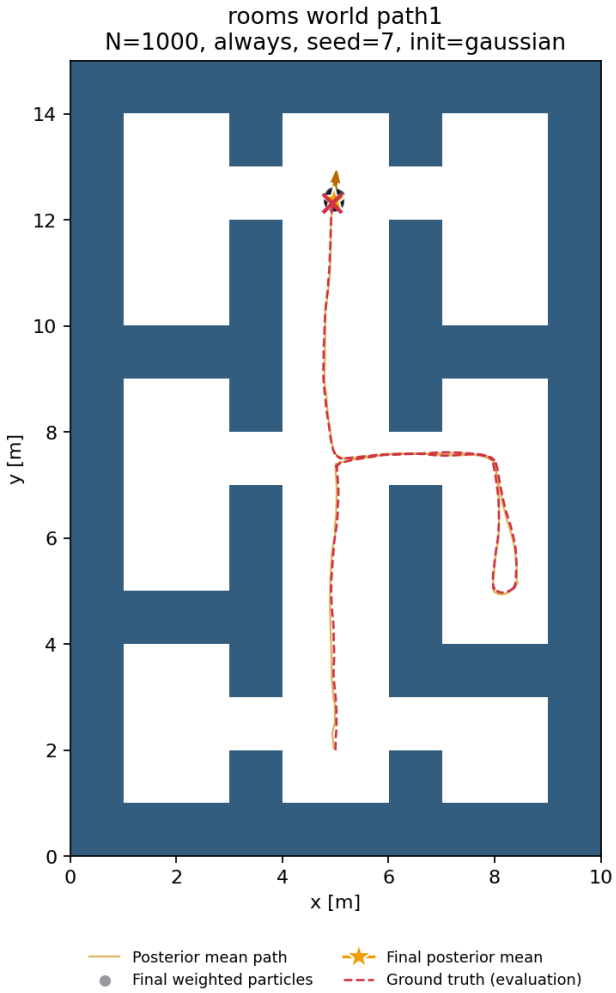
QUESTION 2(D): KNOWN INITIAL POSITION

rooms world path1

Run	Position RMSE	Final position error	Heading RMSE
Failure example: N=10	5.395 m	7.705 m	78.02 deg
Selected: N=1000	0.062 m	0.061 m	2.28 deg



Failure example, N=10, seed 7.



Selected result, N=1000, seed 7.

The ten-particle run follows an incorrect room/corridor hypothesis and repeatedly falls back after occupied motion proposals. The 1000-particle result remains close to the reference trajectory and ends near the true pose.

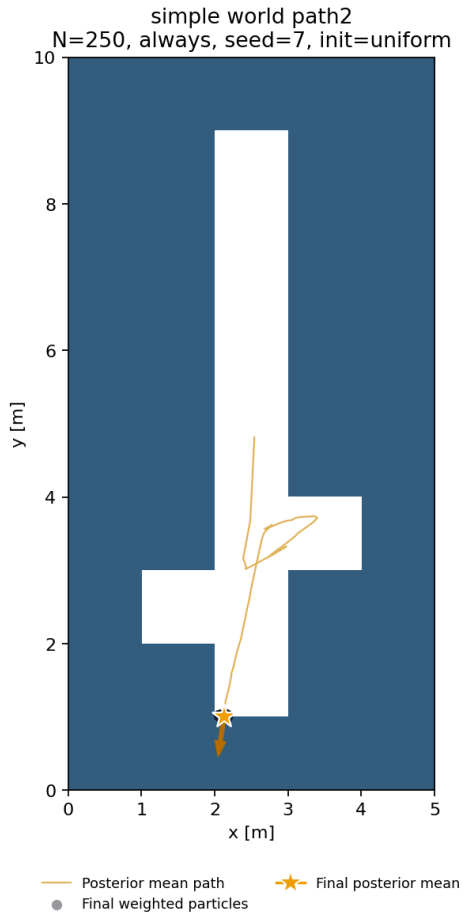
Black points show the final weighted particle population; the orange star is its weighted mean. The dashed red path and red endpoint are ground truth used only for evaluation.

Source: results/pf/rooms_world_path1_N*_always_s7/ (JSON metrics, PNG figures and NPZ histories).

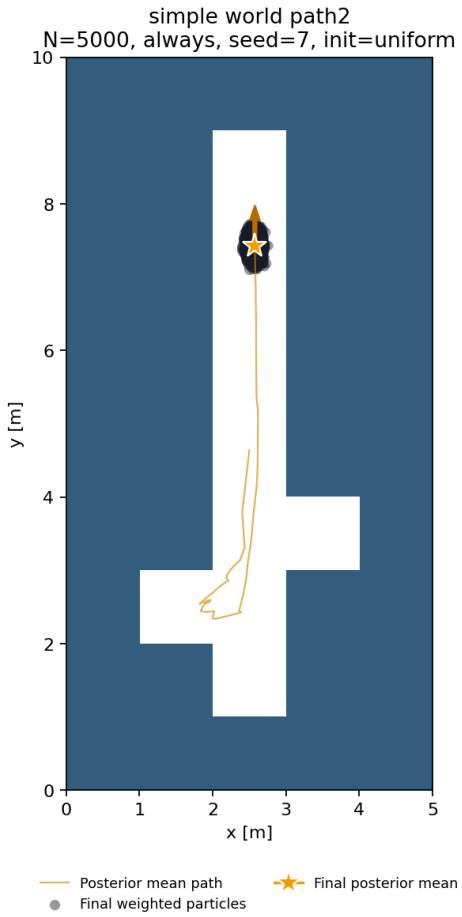
QUESTION 2(E): GLOBAL LOCALIZATION

simple world path2

N	Final scan log L	Spread (m)	Final xy (m)	Fallbacks
250	-7.811	0.028	2.126, 1.010	6382
2000	-7.985	0.018	2.151, 1.013	51094
5000	-1.672	0.101	2.572, 7.430	1



Comparison: N=250.



Selected candidate: N=5000.

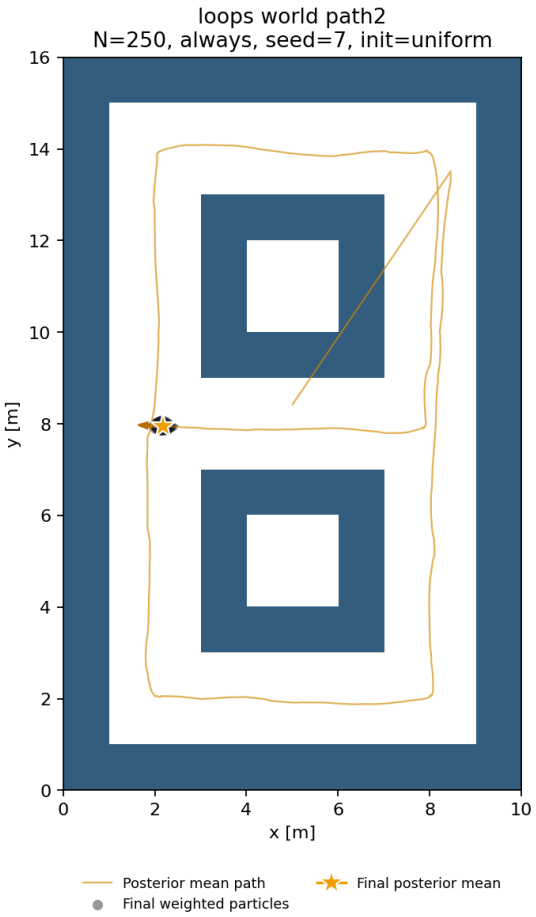
The N=250 and N=2000 runs settle near the lower boundary, with final scan log likelihoods -7.81 and -7.99 and collision fallback rates near 19.5% of particle-steps. N=5000 gives a much better final scan fit (-1.67), a plausible free-space endpoint and only one collision fallback. It is the best supported candidate in this sweep; accuracy is not measurable without ground truth.

The orange path is a history of posterior means. During multimodality its connecting segments can pass through obstacles even when particles lie in free space. No ground-truth accuracy claim is made for this dataset.

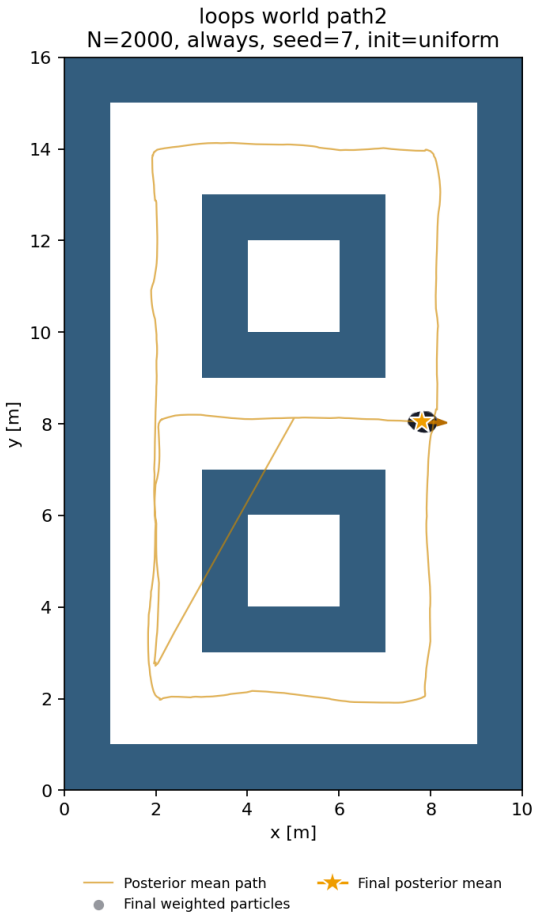
QUESTION 2(E): GLOBAL LOCALIZATION

loops world path2

N	Final scan log L	Spread (m)	Final xy (m)	Fallbacks
250	-2.383	0.067	2.186, 7.948	3
2000	-2.397	0.073	7.807, 8.046	29
5000	-2.388	0.073	7.808, 8.043	91



Comparison: N=250.



Selected candidate: N=2000.

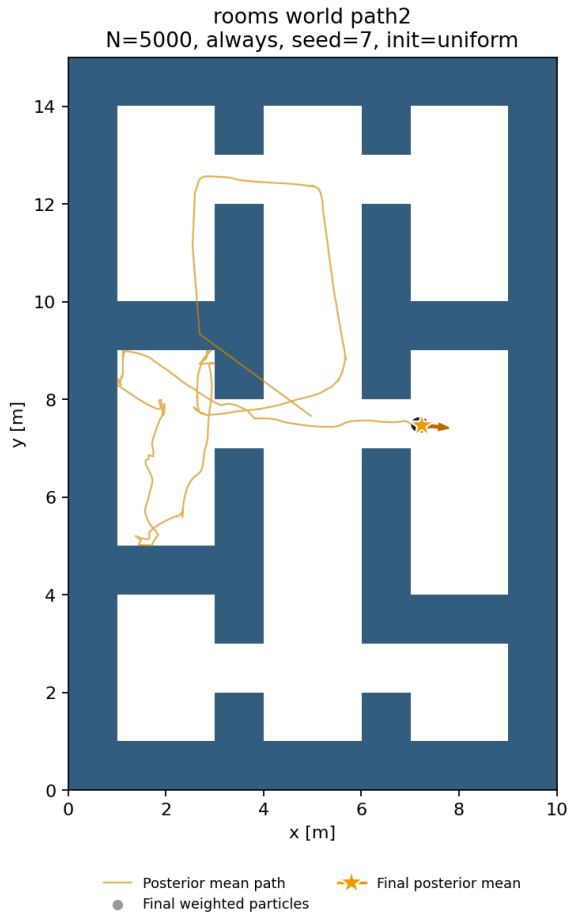
The occupancy map is exactly invariant under a 180-degree rotation about (5,8). Transforming the N=250 pose by (x,y,heading) -> (10-x,16-y,heading+pi) puts it only 0.0094 m and 0.0015 rad from the N=2000 pose. Their likelihoods are nearly equal. These are symmetry-equivalent modes, not evidence that N=250 is wrong. N=2000 and N=5000 end at essentially the same mode; N=2000 is selected as a representative economical setting.

The orange path is a history of posterior means. During multimodality its connecting segments can pass through obstacles even when particles lie in free space. No ground-truth accuracy claim is made for this dataset.

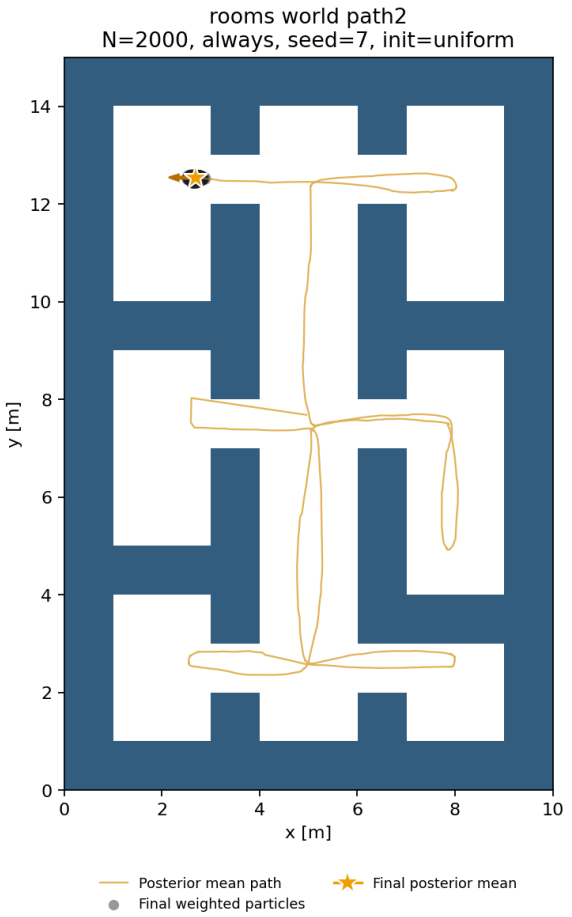
QUESTION 2(E): GLOBAL LOCALIZATION

rooms world path2

N	Final scan log L	Spread (m)	Final xy (m)	Fallbacks
250	-36.740	0.013	1.004, 11.674	18479
2000	-2.151	0.048	2.690, 12.528	55
5000	-7.095	0.036	7.245, 7.471	1265763



Comparison: N=5000.



Selected candidate: N=2000.

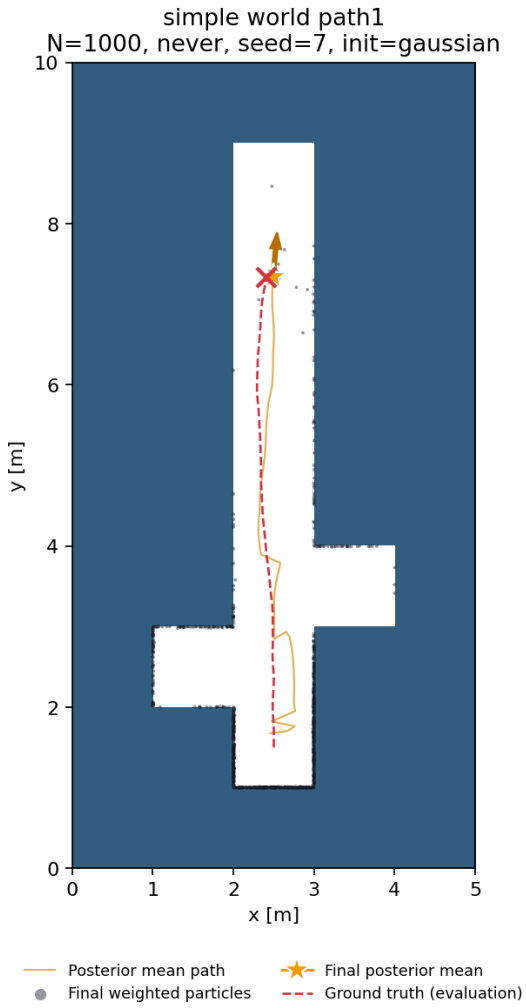
N=2000 provides the strongest final scan fit (-2.15) and only 55 collision fallbacks. N=5000 has a worse fit (-7.09) and 1,265,763 fallbacks, about 24% of particle-steps. A larger count can select a worse hypothesis in a single stochastic realization. N=250 is tightly concentrated with ESS/N=0.975 but has poor scan fit (-36.74); concentration alone is insufficient evidence of success.

The orange path is a history of posterior means. During multimodality its connecting segments can pass through obstacles even when particles lie in free space. No ground-truth accuracy claim is made for this dataset.

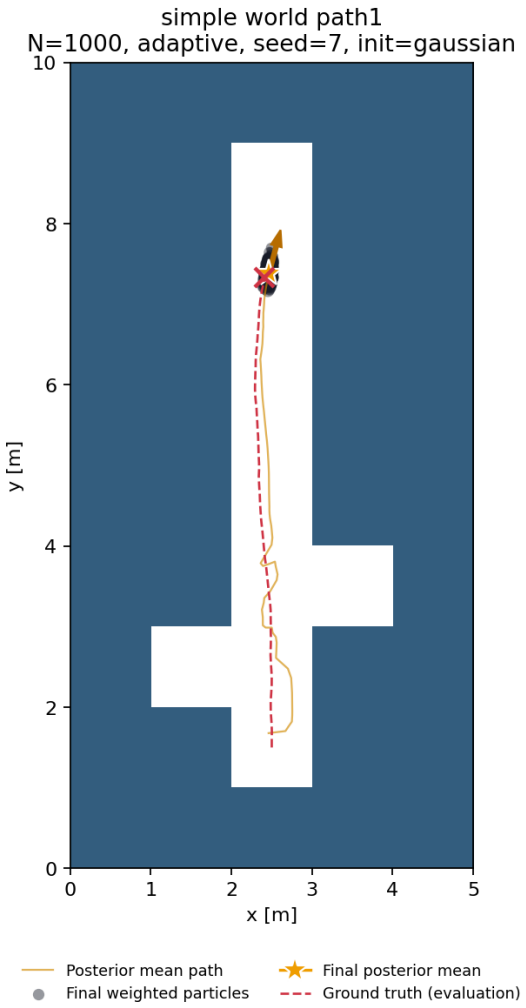
QUESTION 2(F): RESAMPLING COMPARISON

simple world path1 | N=1000

Strategy	RMSE (m)	Final error (m)	Final log L	Final ESS	Resamples
Every step	0.142	0.057	-1.619	916.2	81/81
Never	0.151	0.079	-13.673	1.0	0/81
ESS < 2N/3	0.132	0.078	-1.717	620.2	55/81



Never resample: point sizes still indicate posterior weights.



Adaptive resampling: ESS is measured before resampling.

At the same N, adaptive resampling gives position RMSE 0.132 m, compared with 0.142 m when resampling every step. It resamples 55 of 81 steps. Never-resampling ends with ESS 1.00; its mean is therefore supported by approximately one surviving weighted trajectory.

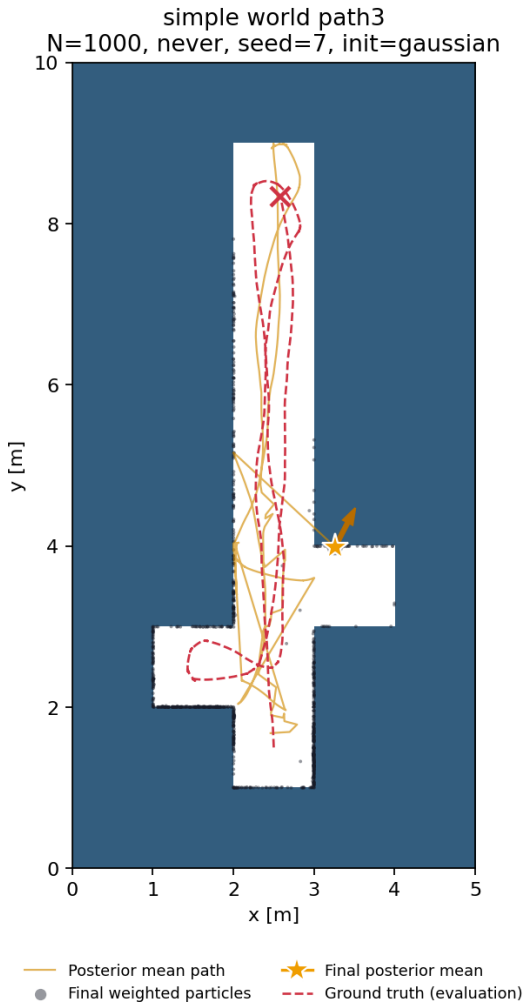
The short simple path retains a small position error without resampling, but its final scan fit and near-unit ESS reveal fragile heading/weight support. Good position error on this short run does not demonstrate a healthy particle approximation.

The corresponding always-resampling figure appears in the earlier selected-count section. All three strategies use the same dataset, seed, initialization rule, sensor settings and particle count; later random draws diverge as resampling changes the population.

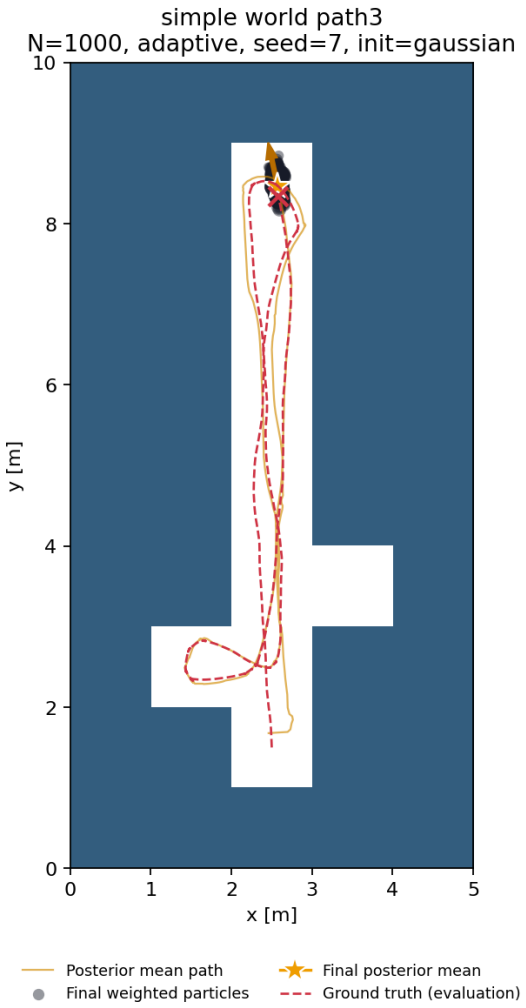
QUESTION 2(F): RESAMPLING COMPARISON

simple world path3 | N=1000

Strategy	RMSE (m)	Final error (m)	Final log L	Final ESS	Resamples
Every step	0.099	0.121	-1.509	874.1	517/517
Never	1.163	4.406	-5.475	1.0	0/517
ESS < 2N/3	0.121	0.132	-1.512	894.7	255/517



Never resample: point sizes still indicate posterior weights.



Adaptive resampling: ESS is measured before resampling.

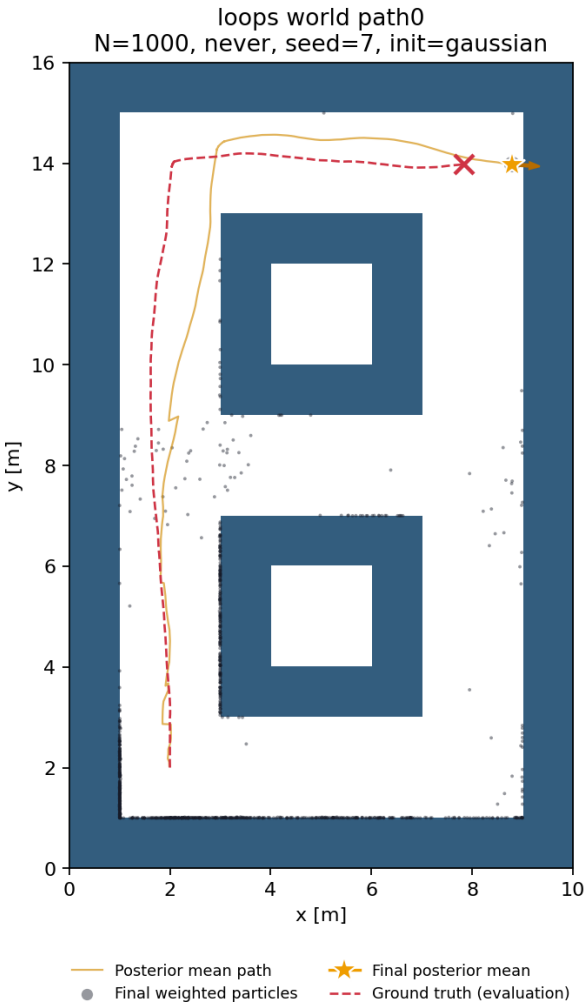
At the same N, adaptive resampling gives position RMSE 0.121 m, compared with 0.099 m when resampling every step. It resamples 255 of 517 steps. Never-resampling ends with ESS 1.00; its mean is therefore supported by approximately one surviving weighted trajectory.

The corresponding always-resampling figure appears in the earlier selected-count section. All three strategies use the same dataset, seed, initialization rule, sensor settings and particle count; later random draws diverge as resampling changes the population.

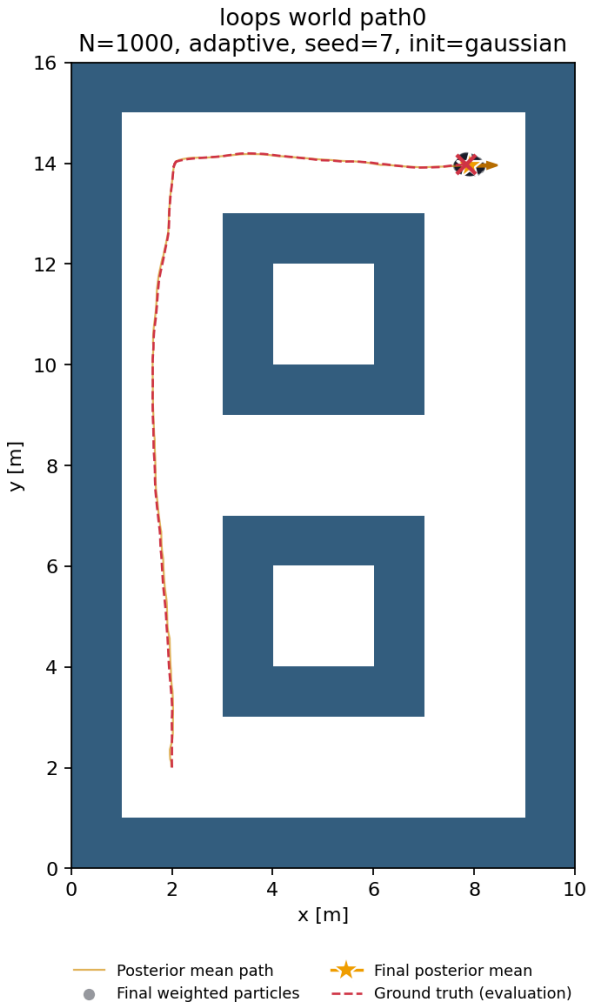
QUESTION 2(F): RESAMPLING COMPARISON

loops world path0 | N=1000

Strategy	RMSE (m)	Final error (m)	Final log L	Final ESS	Resamples
Every step	0.057	0.085	-2.328	812.7	249/249
Never	0.797	0.953	-14.851	1.0	0/249
ESS < 2N/3	0.054	0.076	-2.311	506.4	149/249



Never resample: point sizes still indicate posterior weights.



Adaptive resampling: ESS is measured before resampling.

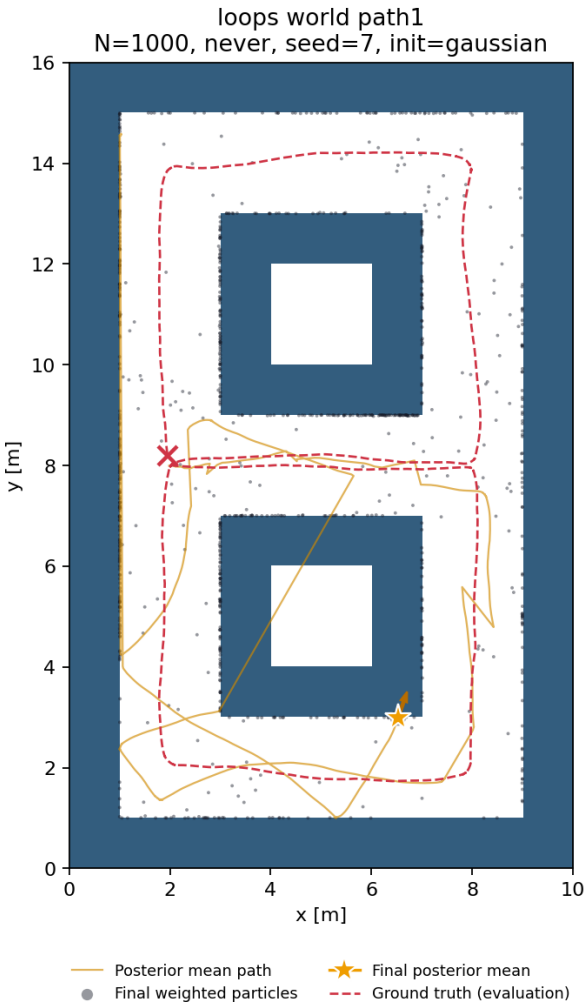
At the same N, adaptive resampling gives position RMSE 0.054 m, compared with 0.057 m when resampling every step. It resamples 149 of 249 steps. Never-resampling ends with ESS 1.00; its mean is therefore supported by approximately one surviving weighted trajectory.

The corresponding always-resampling figure appears in the earlier selected-count section. All three strategies use the same dataset, seed, initialization rule, sensor settings and particle count; later random draws diverge as resampling changes the population.

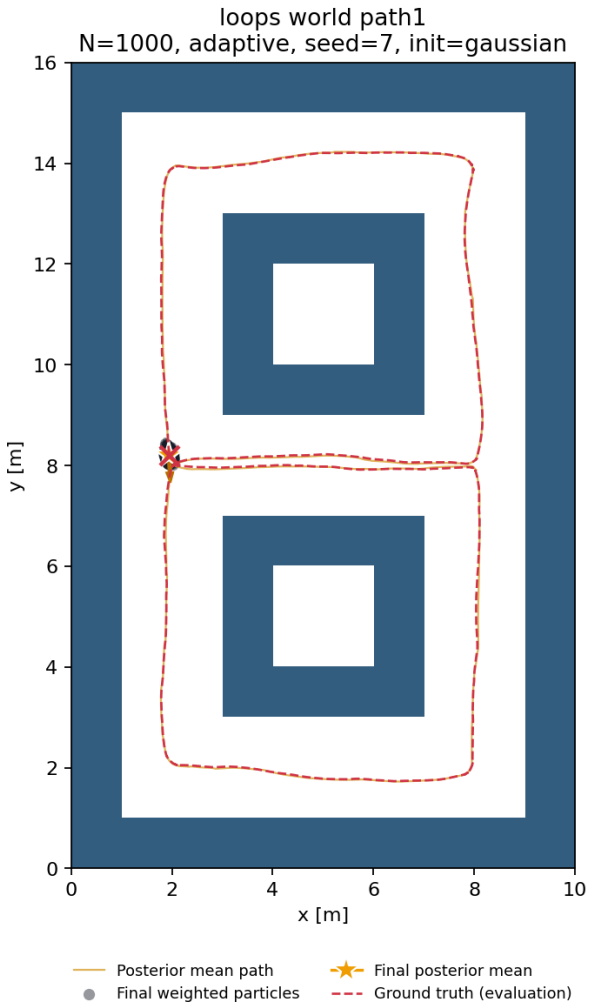
QUESTION 2(F): RESAMPLING COMPARISON

loops world path1 | N=1000

Strategy	RMSE (m)	Final error (m)	Final log L	Final ESS	Resamples
Every step	0.045	0.023	-2.560	901.3	790/790
Never	6.519	6.923	-50.999	1.0	0/790
ESS < 2N/3	0.046	0.028	-2.576	720.4	451/790



Never resample: point sizes still indicate posterior weights.



Adaptive resampling: ESS is measured before resampling.

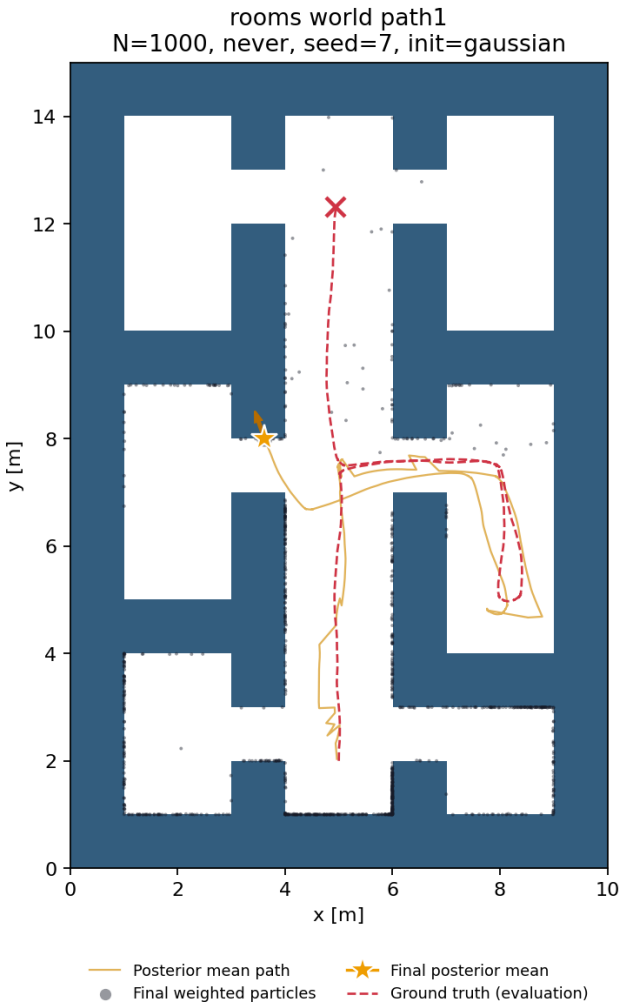
At the same N, adaptive resampling gives position RMSE 0.046 m, compared with 0.045 m when resampling every step. It resamples 451 of 790 steps. Never-resampling ends with ESS 1.00; its mean is therefore supported by approximately one surviving weighted trajectory.

The corresponding always-resampling figure appears in the earlier selected-count section. All three strategies use the same dataset, seed, initialization rule, sensor settings and particle count; later random draws diverge as resampling changes the population.

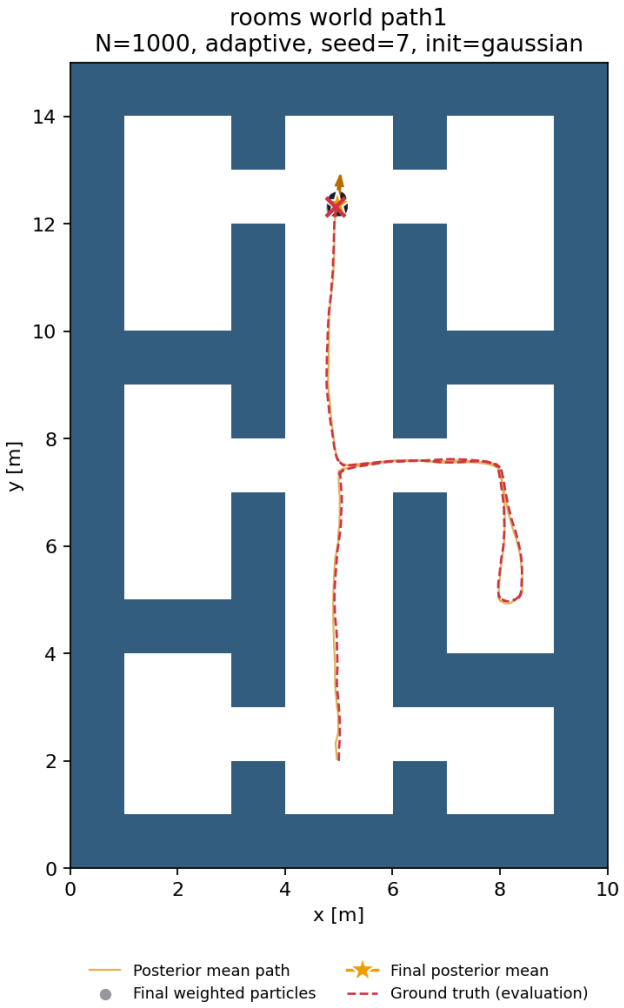
QUESTION 2(F): RESAMPLING COMPARISON

rooms world path1 | N=1000

Strategy	RMSE (m)	Final error (m)	Final log L	Final ESS	Resamples
Every step	0.062	0.061	-2.297	120.0	436/436
Never	1.115	4.518	-40.594	1.0	0/436
ESS < 2N/3	0.063	0.056	-2.307	132.6	222/436



Never resample: point sizes still indicate posterior weights.



Adaptive resampling: ESS is measured before resampling.

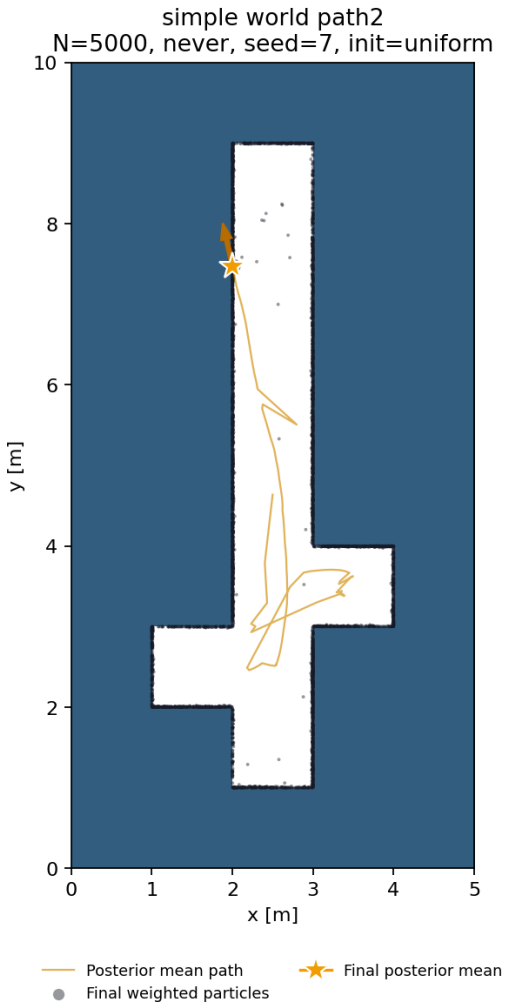
At the same N, adaptive resampling gives position RMSE 0.063 m, compared with 0.062 m when resampling every step. It resamples 222 of 436 steps. Never-resampling ends with ESS 1.00; its mean is therefore supported by approximately one surviving weighted trajectory.

The corresponding always-resampling figure appears in the earlier selected-count section. All three strategies use the same dataset, seed, initialization rule, sensor settings and particle count; later random draws diverge as resampling changes the population.

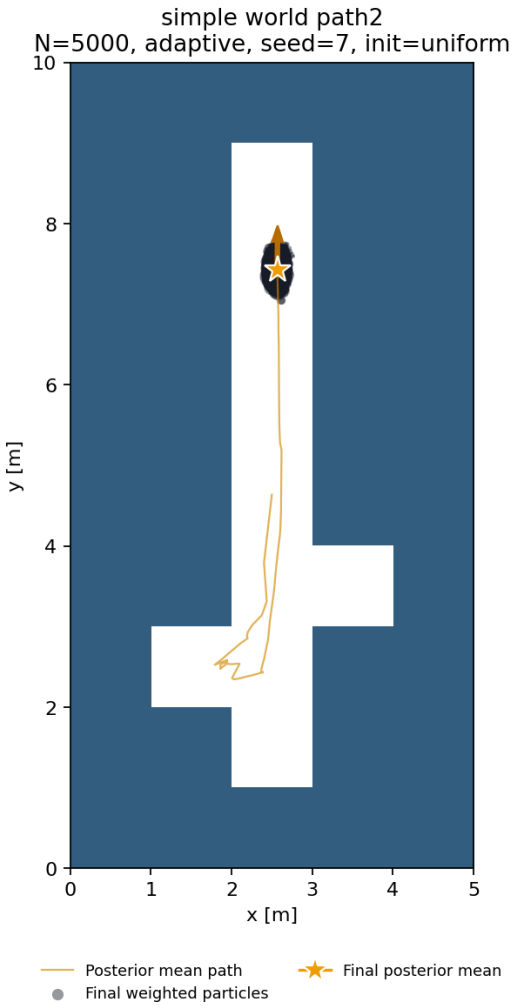
QUESTION 2(F): RESAMPLING COMPARISON

simple world path2 | N=5000

Strategy	RMSE (m)	Final error (m)	Final log L	Final ESS	Resamples
Every step	n/a	n/a	-1.672	4135.3	131/131
Never	n/a	n/a	-23.006	1.0	0/131
ESS < 2N/3	n/a	n/a	-1.661	3338.3	64/131



Never resample: point sizes still indicate posterior weights.



Adaptive resampling: ESS is measured before resampling.

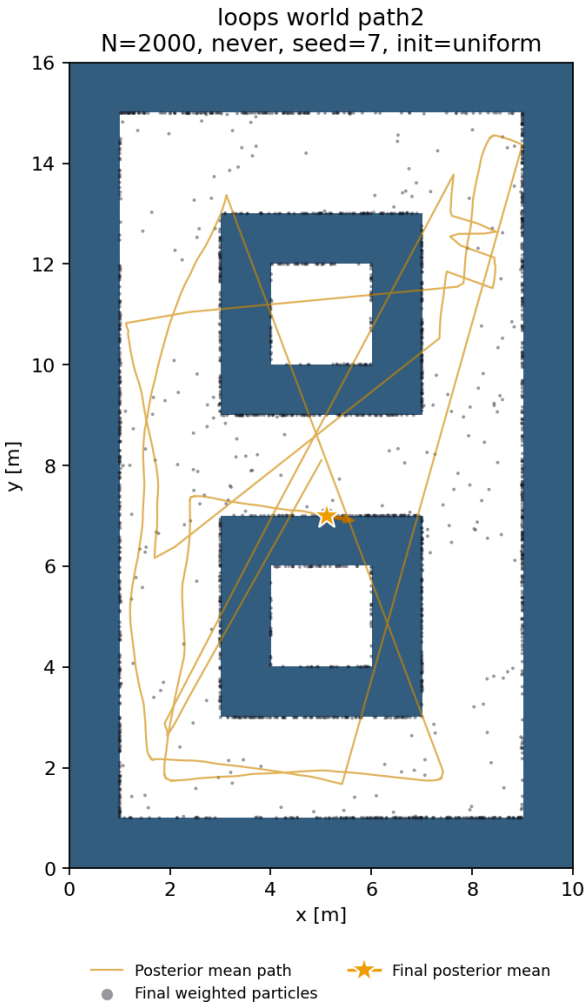
Without resampling, final scan log L is -23.006, versus -1.661 with ESS-triggered resampling. The adaptive result preserves the selected run's scan consistency, while the never-resampled weights collapse to nearly one trajectory. These are scan-fit comparisons, not accuracy measurements.

The corresponding always-resampling figure appears in the earlier selected-count section. All three strategies use the same dataset, seed, initialization rule, sensor settings and particle count; later random draws diverge as resampling changes the population.

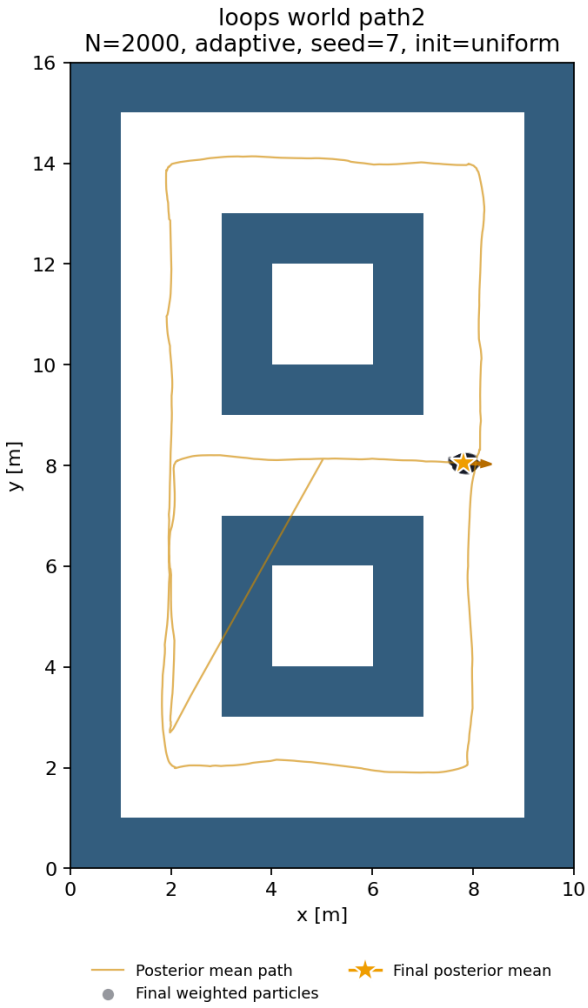
QUESTION 2(F): RESAMPLING COMPARISON

loops world path2 | N=2000

Strategy	RMSE (m)	Final error (m)	Final log L	Final ESS	Resamples
Every step	n/a	n/a	-2.397	1837.5	731/731
Never	n/a	n/a	-49.700	1.0	0/731
ESS < 2N/3	n/a	n/a	-2.377	1207.3	403/731



Never resample: point sizes still indicate posterior weights.



Adaptive resampling: ESS is measured before resampling.

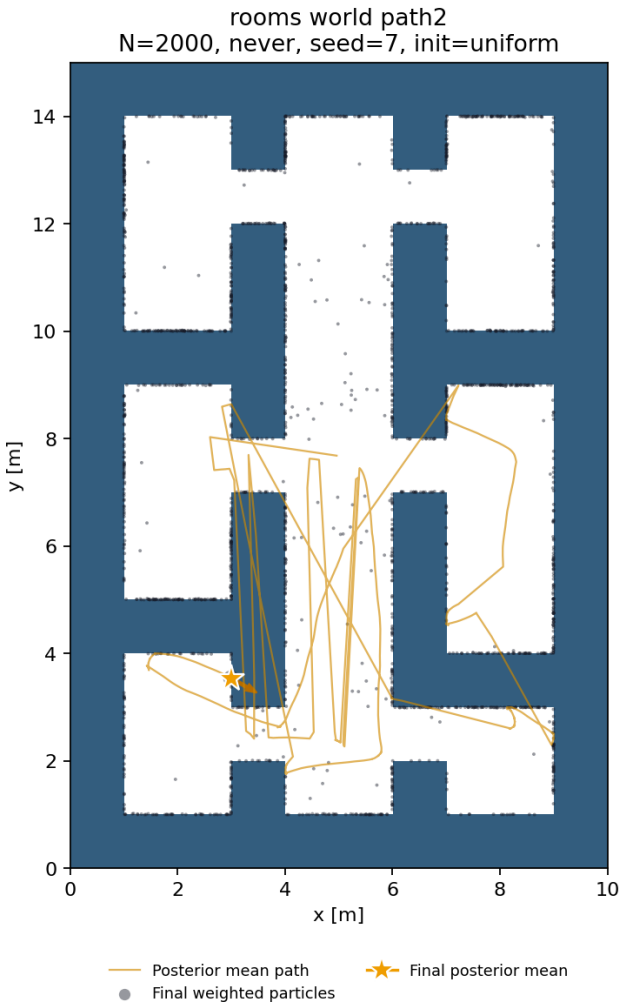
Without resampling, final scan log L is -49.700, versus -2.377 with ESS-triggered resampling. The adaptive result preserves the selected run's scan consistency, while the never-resampled weights collapse to nearly one trajectory. These are scan-fit comparisons, not accuracy measurements.

The corresponding always-resampling figure appears in the earlier selected-count section. All three strategies use the same dataset, seed, initialization rule, sensor settings and particle count; later random draws diverge as resampling changes the population.

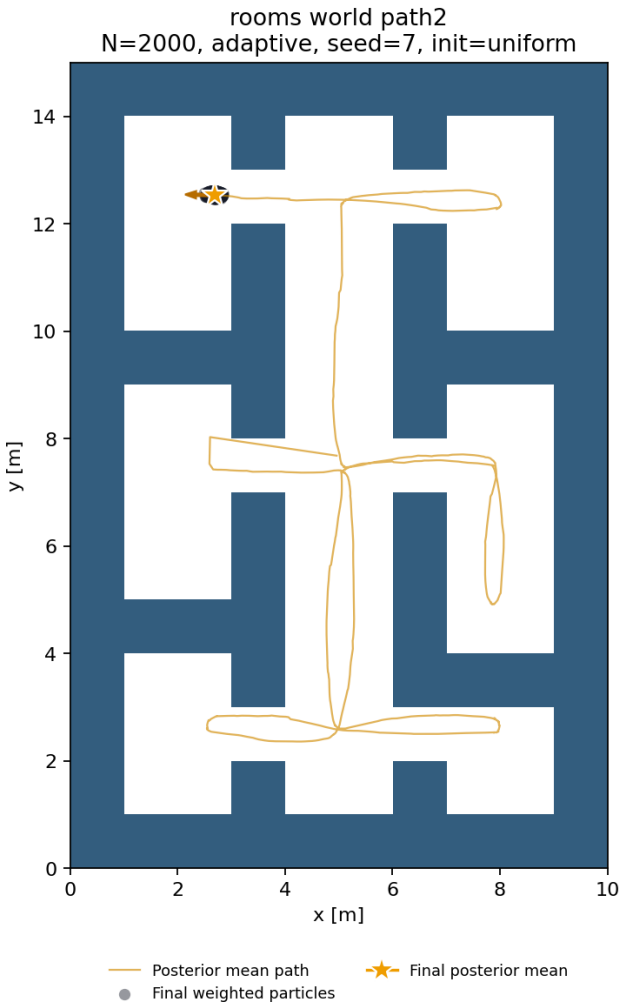
QUESTION 2(F): RESAMPLING COMPARISON

rooms world path2 | N=2000

Strategy	RMSE (m)	Final error (m)	Final log L	Final ESS	Resamples
Every step	n/a	n/a	-2.151	1911.2	1053/1053
Never	n/a	n/a	-50.136	1.0	0/1053
ESS < 2N/3	n/a	n/a	-2.152	1710.4	594/1053



Never resample: point sizes still indicate posterior weights.



Adaptive resampling: ESS is measured before resampling.

Without resampling, final scan log L is -50.136, versus -2.152 with ESS-triggered resampling. The adaptive result preserves the selected run's scan consistency, while the never-resampled weights collapse to nearly one trajectory. These are scan-fit comparisons, not accuracy measurements.

The corresponding always-resampling figure appears in the earlier selected-count section. All three strategies use the same dataset, seed, initialization rule, sensor settings and particle count; later random draws diverge as resampling changes the population.

QUESTION 2(F): INCREASED-COUNT EXPERIMENTS

Can more particles rescue never-resampling?

Seven additional completed runs increase N to 10,000 for the longer known-start cases and 20,000 for global localization. These are bounded follow-up experiments; they do not establish the minimum count needed for convergence.

Dataset	N	RMSE m	Final m	Tail20 m	Log L	ESS
simple path3	10000	0.350	0.757	0.473	-6.764	1.00
loops path0	10000	0.386	0.245	0.322	-9.973	1.00
loops path1	10000	1.232	3.880	2.227	-46.918	1.00
rooms path1	10000	0.564	0.993	0.678	-42.856	1.00
simple path2	20000	n/a	n/a	n/a	-2.705	1.00
loops path2	20000	n/a	n/a	n/a	-16.282	1.00
rooms path2	20000	n/a	n/a	n/a	-30.846	1.00

Exploratory position criterion. For known-start recovery, require both final position error < 0.5 m and tail-20% position RMSE < 0.5 m. This is a practical criterion for these experiments, not a course-prescribed threshold or a test of full-posterior reliability. A small terminal error alone can occur by chance after substantial earlier error.

Known-start outcomes. At N=10,000, loops_world_path0 meets that position criterion (final error 0.245 m, tail RMSE 0.322 m), although ESS remains one. simple_world_path3 fails because its final error is 0.757 m; loops_world_path1 and rooms_world_path1 also fail. No successful count is claimed for those cases within the tested range.

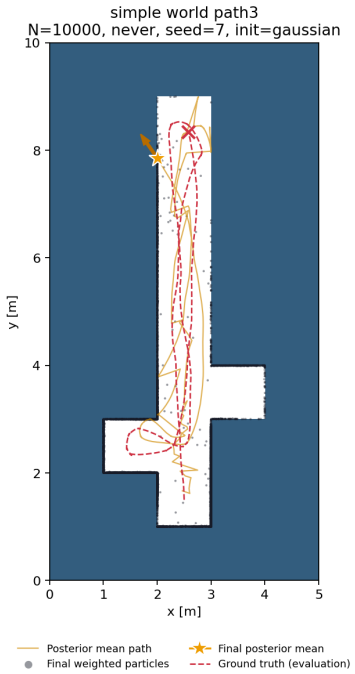
Unknown-start outcomes. At N=20,000, simple_world_path2 improves its final scan log likelihood to -2.705, but ESS remains one and no ground truth is available. The increased counts do not restore adequate scan consistency for loops_world_path2 (-16.282) or rooms_world_path2 (-30.846), compared with their selected resampling baselines near -2.4 and -2.15. No tested count cures those failures; a successful threshold was not established.

A near-zero weighted spread is not reassuring when ESS is one: it reflects numerical concentration of the weights, not evidence that the pose is known exactly. The full particle arrays and log-weight-derived normalized weights are saved with each run.

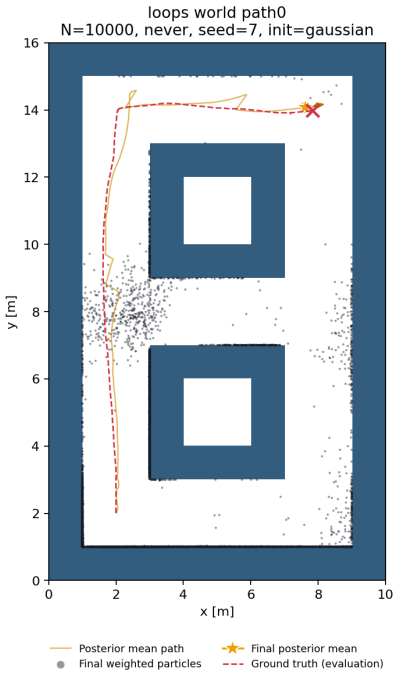
The next two pages include every increased-count final-distribution figure. Original full-size PNG files and the exact commands are in results/pf/<run_id>/; execution.json records successful completion, and the neighboring JSON/NPZ files contain metrics and histories.

QUESTION 2(F): SUPPLEMENTARY FIGURES

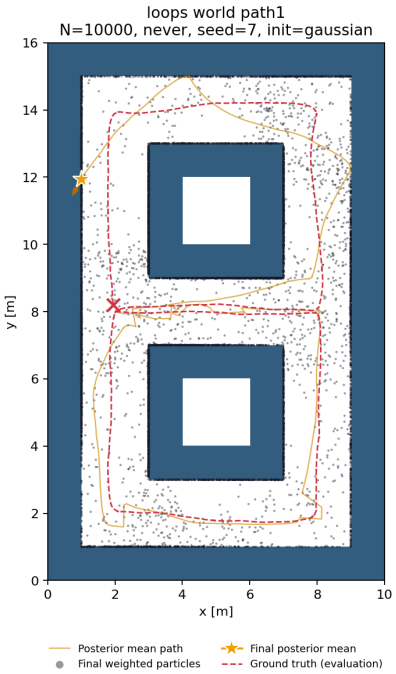
Increased-count final distributions



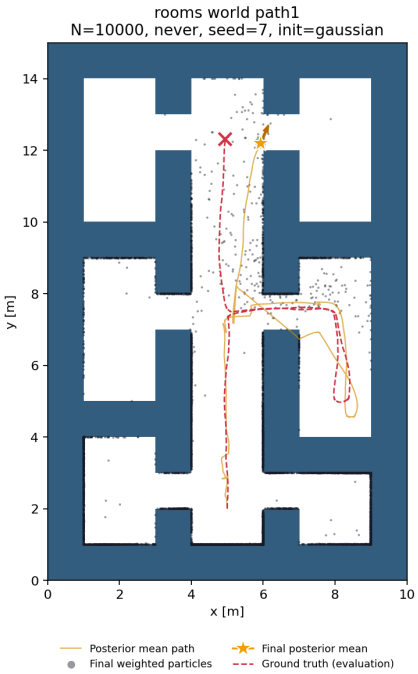
simple_world_path3, N=10000, never, seed 7.



loops_world_path0, N=10000, never, seed 7.



loops_world_path1, N=10000, never, seed 7.

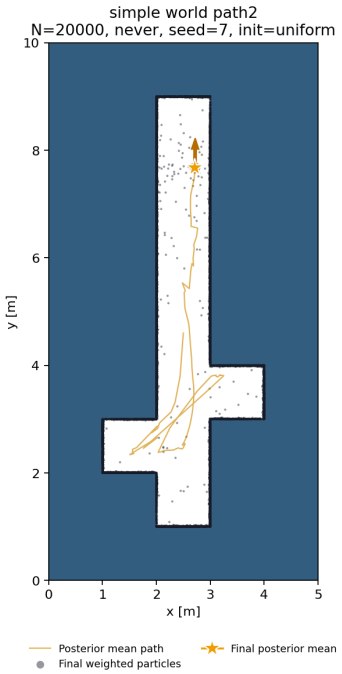


rooms_world_path1, N=10000, never, seed 7.

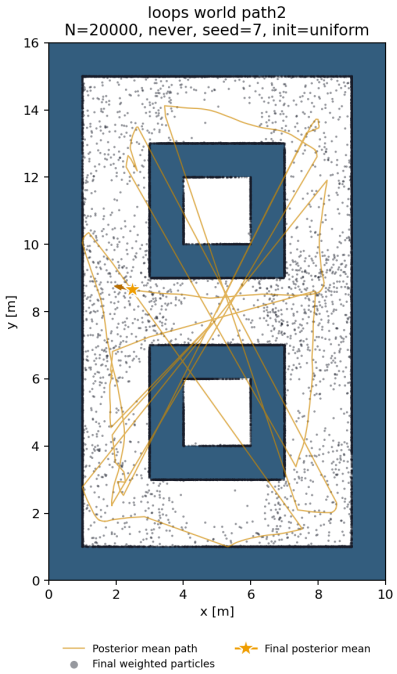
Particle marker sizes encode normalized weights. An apparently broad raw population can still have approximately one effective particle. Consult the preceding table for ESS and numerical performance.

QUESTION 2(F): SUPPLEMENTARY FIGURES

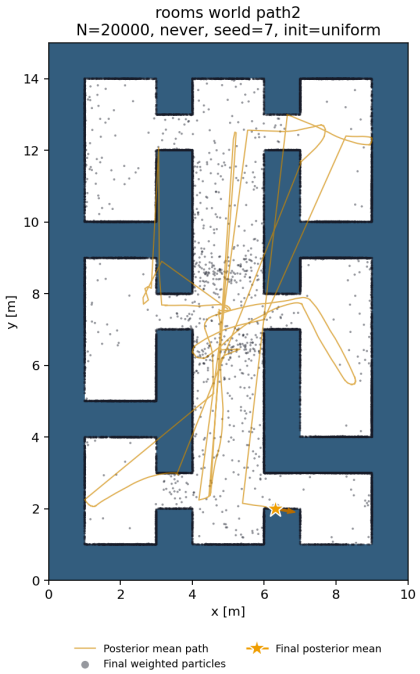
Increased-count final distributions



simple_world_path2, N=20000, never, seed 7.



loops_world_path2, N=20000, never, seed 7.



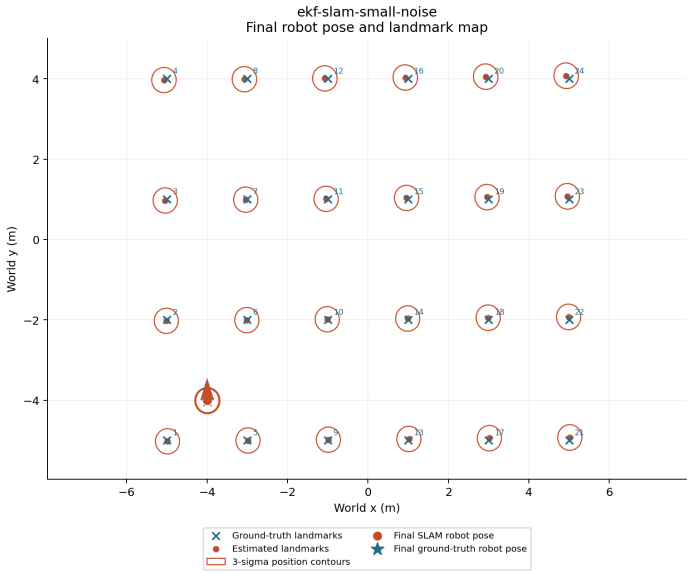
rooms_world_path2, N=20000, never, seed 7.

Particle marker sizes encode normalized weights. An apparently broad raw population can still have approximately one effective particle. Consult the preceding table for ESS and numerical performance.

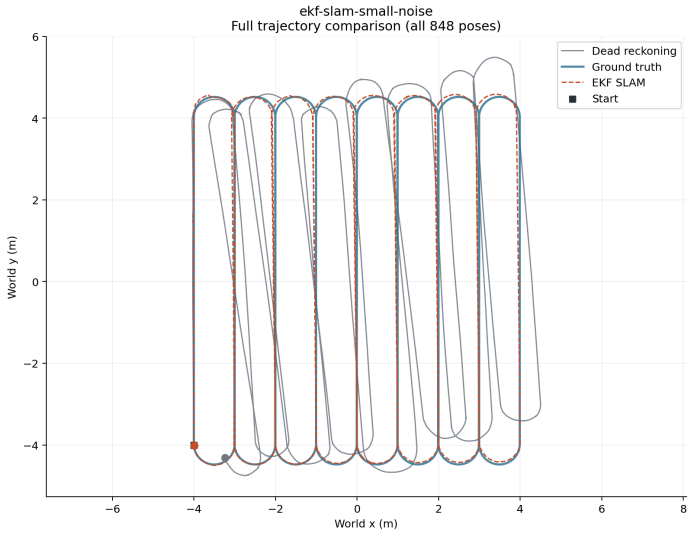
QUESTION 3(B): JOINT POSE AND LANDMARK ESTIMATION

EKF SLAM: small-noise dataset

Metric	EKF SLAM	Dead reckoning
Position RMSE (m)	0.067	0.646
Final position error (m)	0.013	0.825
Heading RMSE (degrees)	0.641	5.321
Map position RMSE (m)	0.060	No map estimated



Final pose and landmark estimates with 3-sigma covariance contours and ground truth.



SLAM, dead-reckoned and ground-truth trajectories.

The filter processes 847 controls and 1473 observations, initializes all 24 landmarks, and performs 1449 updates on already-known landmarks. Initial pose comes from X0. New observations augment the state once; that same observation is not immediately reused for a second update.

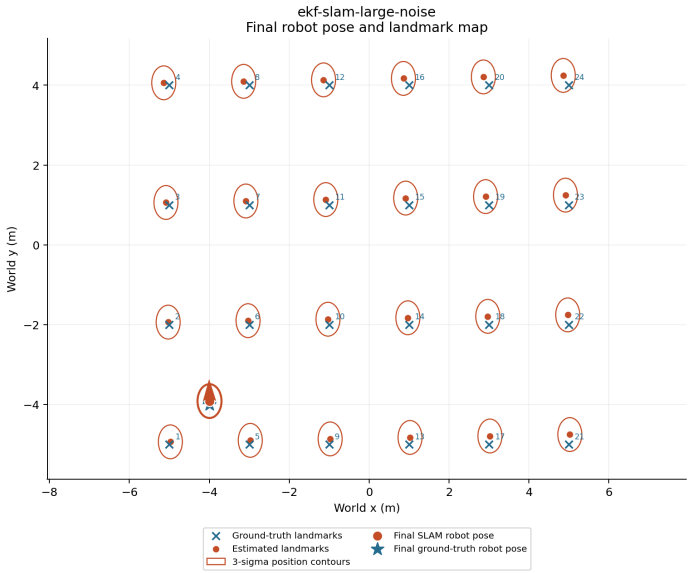
Prediction uses the supplied two-noise distance/heading model and propagates robot-map cross-covariance. Cartesian observation Jacobians update the entire joint state, using a linear solve for the gain and Joseph-form covariance. Landmark initialization rotates the relative Cartesian reading into world coordinates and retains all induced correlations.

Saved R diagonal: 0.001, 3.49037e-05. Saved Q diagonal: 1e-05, 1e-05. The file named 'large-noise' has about ten times larger process variances but ten times smaller measurement variances than the 'small-noise' file. The filenames do not mean that every noise source is larger.

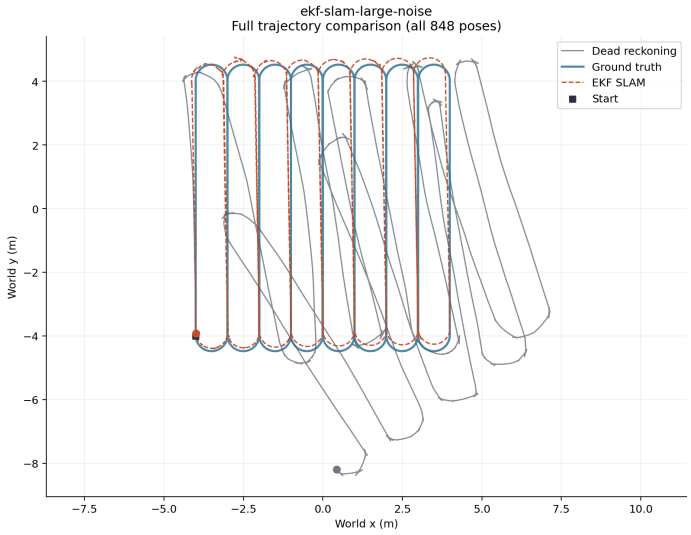
QUESTION 3(B): JOINT POSE AND LANDMARK ESTIMATION

EKF SLAM: large-noise dataset

Metric	EKF SLAM	Dead reckoning
Position RMSE (m)	0.190	2.883
Final position error (m)	0.084	6.108
Heading RMSE (degrees)	1.264	18.560
Map position RMSE (m)	0.186	No map estimated



Final pose and landmark estimates with 3-sigma covariance contours and ground truth.



SLAM, dead-reckoned and ground-truth trajectories.

The filter processes 847 controls and 1462 observations, initializes all 24 landmarks, and performs 1438 updates on already-known landmarks. Initial pose comes from X0. New observations augment the state once; that same observation is not immediately reused for a second update.

Prediction uses the supplied two-noise distance/heading model and propagates robot-map cross-covariance. Cartesian observation Jacobians update the entire joint state, using a linear solve for the gain and Joseph-form covariance. Landmark initialization rotates the relative Cartesian reading into world coordinates and retains all induced correlations.

Saved R diagonal: 0.01, 0.000349083. Saved Q diagonal: 1e-06, 1e-06. The file named 'large-noise' has about ten times larger process variances but ten times smaller measurement variances than the 'small-noise' file. The filenames do not mean that every noise source is larger.

SUPPORTING INFORMATION

Verification, reproducibility and accounting

34 analytic tests pass: 12 particle-filter tests, 11 LIDAR tests and 11 EKF tests. The test code is in tests/ and uses unittest. It covers zero/near-zero turns, angular wrapping, collisions, noise variances, recursive and log-domain weights, ESS/resampling, beam-model normalization and support, physical ray distances, and finite-difference checks of full joint EKF covariance calculations.

The EKF tests independently check known-landmark updates and augmentation, correlated process noise, initial-time observations, prediction-before-measurement ordering, no double-use of first observations, and independence from supplied ground-truth arrays. Both complete EKF runs retain finite states, symmetric covariance and positive covariance eigenvalues at recorded posteriors.

Task	Command from the submission directory
Run analytic tests	python3 -m unittest discover -s tests -v
Run particle filter	python3 code/pf/RunPF.py data/pf/simple_world_path1.pickle --numParticles 1000 --sparsity 5 --seed 7
Run EKF SLAM	python3 code/ekf-slam/RunEKFSLAM.py data/ekf-slam/ekf-slam-small-noise.pickle
Rebuild this report	python3 reports/build_report.py

Evidence files. Each particle-filter run has execution.json, stdout/stderr, a metrics JSON, a PNG final-cloud plot and an NPZ with particles, weights, estimates, ESS and errors. EKF directories contain metrics.json, results.npz, final_map and trajectories figures. Experiment manifests record the selected settings and follow-up runs. The report reads those saved files and does not alter or rerun inference.

Question 4 / honest accounting. Human collaborators: enter the actual names, or 'none' if accurate. Time spent: enter your actual time; it cannot be inferred from this report. AI assistance: ChatGPT was used to explain concepts, derive formulas, implement and review the requested code, construct analytic tests, run experiments and prepare this report. Review the implementation and results before submission, and disclose assistance as required by the course.

References.

[1] TTIC 31170, Robot Learning and Estimation, Autumn 2026, Problem Set 3: Questions 2 and 3, supplied ps3-questions.pdf.
[2] Supplied Python starter modules and README files in code/pf, code/ekf-slam and data.
[3] S. Thrun, W. Burgard and D. Fox, Probabilistic Robotics, MIT Press, 2005. Motion and beam algorithms: Tables 5.3 and 6.1, as referenced by the assignment.
[4] The saved experiment metrics, figures and histories bundled with this submission.